

Investigating Robust Overfitting in Adversarial Training

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1. Explaining and Harnessing Adversarial Examples

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EXPLAINING AND HARNESSING ADVERSARIAL EXAMPLES

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ABSTRACT

Several machine learning models, including neural networks, consistently misclassify *adversarial examples*—inputs formed by applying small but intentionally worst-case perturbations to examples from the dataset, such that the perturbed input results in the model outputting an incorrect answer with high confidence. Early attempts at explaining this phenomenon focused on nonlinearity and overfitting. We argue instead that the primary cause of neural networks' vulnerability to adversarial perturbation is their linear nature. This explanation is supported by new quantitative results while giving the first explanation of the most intriguing fact about them: their generalization across architectures and training sets. Moreover, this view yields a simple and fast method of generating adversarial examples. Using this approach to provide examples for adversarial training, we reduce the test set error of a maxout network on the MNIST dataset.

1 INTRODUCTION

Szegedy et al. (2014b) made an intriguing discovery: several machine learning models, including state-of-the-art neural networks, are vulnerable to *adversarial examples*. That is, these machine learning models misclassify examples that are only slightly different from correctly classified examples drawn from the data distribution. In many cases, a wide variety of models with different architectures trained on different subsets of the training data misclassify the same adversarial example. This suggests that adversarial examples expose fundamental blind spots in our training algorithms.

The cause of these adversarial examples was a mystery, and speculative explanations have suggested it is due to extreme nonlinearity of deep neural networks, perhaps combined with insufficient model averaging and insufficient regularization of the purely supervised learning problem. We show that these speculative hypotheses are unnecessary. Linear behavior in high-dimensional spaces is sufficient to cause adversarial examples. This view enables us to design a fast method of generating adversarial examples that makes adversarial training practical. We show that adversarial training can provide an additional regularization benefit beyond that provided by using dropout (Srivastava et al., 2014) alone. Generic regularization strategies such as dropout, pretraining, and model averaging do not confer a significant reduction in a model's vulnerability to adversarial examples, but changing to nonlinear model families such as RBF networks can do so.

Our explanation suggests a fundamental tension between designing models that are easy to train due to their linearity and designing models that use nonlinear effects to resist adversarial perturbation. In the long run, it may be possible to escape this tradeoff by designing more powerful optimization methods that can successfully train more nonlinear models.

2 RELATED WORK

Szegedy et al. (2014b) demonstrated a variety of intriguing properties of neural networks and related models. Those most relevant to this paper include:

- Box-constrained L-BFGS can reliably find adversarial examples.
- On some datasets, such as ImageNet (Deng et al., 2009), the adversarial examples were so close to the original examples that the differences were indistinguishable to the human eye.
- The same adversarial example is often misclassified by a variety of classifiers with different architectures or trained on different subsets of the training data.

Cause

Common Belief:

- Data scientists thought adversarial examples happened because deep networks were too complex and nonlinear. [Expand this]
- The idea was that networks were so complicated and overfit to detail that unusual inputs could cause errors the model was not prepared for.

What this paper found instead:

- Modern networks are built to act linearly on purpose, since linear behavior makes training much easier.
- Common building blocks like ReLUs, LSTMs, and Maxout Networks all lean toward linear behavior.
- That same linear behavior is the vulnerability, not complexity. [Explain what this means]

How These Changes Contribute

- Images exist in very high dimensional spaces. Even a small 100x100 color image has 30,000 input values.
- An attacker can change every single pixel by a tiny, nearly invisible amount.
- In a linear model, thousands of tiny changes in the same direction combine into one large change in the output.
- That combined change is enough to turn a confident, correct answer into a confident, wrong one.
- Humans do not have strong intuition for this because we only experience 3 dimensions.

Example:

- Adding a tiny calculated noise pattern (epsilon = 0.007) causes a model to classify a panda image as a gibbon.
- Original panda classification: 57.7% confidence. After the noise is added: 99.3% confidence that the image is a gibbon.

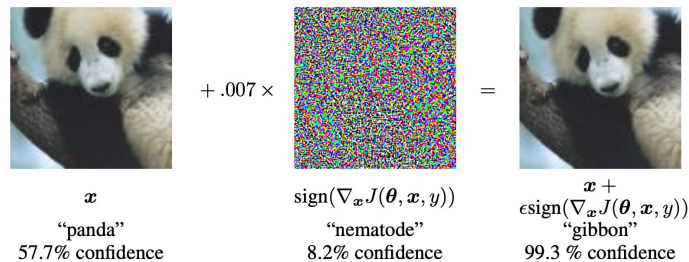


Figure 1: A demonstration of fast adversarial example generation applied to GoogLeNet (Szegedy et al., 2014a) on ImageNet. By adding an imperceptibly small vector whose elements are equal to the sign of the elements of the gradient of the cost function with respect to the input, we can change GoogLeNet's classification of the image. Here our ϵ of .007 corresponds to the magnitude of the smallest bit of an 8 bit image encoding after GoogLeNet's conversion to real numbers.

Fast Gradient Sign Method (FGSM)

- FGSM is a fast, one-step method for creating adversarial examples.
- It uses backpropagation, the same calculation a network uses to learn, but applies it in the direction that increases the model's error instead of the direction that reduces it.
- Formula idea: take the sign of the gradient of the loss, and change the input by a small amount in that direction.
- This approximation works well for neural networks but is only the exact worst-case attack for simpler models like logistic regression.

$$\eta = \epsilon \operatorname{sign}(\nabla_x J(\theta, x, y))$$

Why this matters for us:

- FGSM is the direct predecessor to PGD, the attack Rice et al. use throughout their experiments.
- PGD repeats FGSM's one-step method multiple times using small steps.
- Understanding FGSM first is necessary background before PGD's multi-step version can be understood.

Early Observation of Robust Overfitting

- Adversarial training (adding adversarial examples into training data) reduced error on adversarial examples from 89.4% down to 17.9% on MNIST.
- Caveat: even at 17.9% error, the model's average confidence on its wrong answers was still 81.4%. It made errors less often, but with equally high confidence.
- Early observation of our exact research problem: when the authors scaled up their network, clean validation error stayed flat during training, but adversarial validation error did not. They had to switch to early stopping based on adversarial validation error specifically.
- This is the same separation between clean and robust performance that Rice et al. later define and study in depth as robust overfitting.
- Broader connection: if models memorize perturbation directions instead of learning a genuinely robust boundary, that is exactly what this paper's linearity argument predicts linear models are likely to do.

Bonus: Rubbish Class Examples

- Pure noise, with no real object in it, can still fool a model.
- Basic test: authors fed 10,000 samples of random Gaussian noise directly into classifiers, with no modification. A naively trained maxout network classified 98.35% of these as real objects, with 92.8% average confidence on the wrong answers.
- Targeted test: authors took a Gaussian noise sample and applied a single gradient step toward one specific category, such as "airplane." To a human, the image still looks like colorful static. The network classifies it as an airplane with over 50% confidence.
- Why this matters: linear models produce more extreme, overconfident predictions the farther a point is from real training data. Random noise is very far from real training data, so a linear model still assigns it a confident, extreme label instead of admitting uncertainty.
- RBF networks do not have this problem. They achieve 0% error on rubbish examples, because they only respond confidently to points near training data. Everywhere else, they stay uncertain.
- Connection to our research: this is the same linearity argument from the earlier slides, now applied to random noise instead of adversarial perturbations. It reinforces that linear behavior, not model complexity, drives overconfident mistakes

Overfitting in adversarially robust deep learning

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Abstract

It is common practice in deep learning to use over-parameterized networks and train for as long as possible; there are numerous studies that show, both theoretically and empirically, that such practices surprisingly do not unduly harm the generalization performance of the classifier. In this paper, we empirically study this phenomenon in the setting of *adversarially trained deep networks*, which are trained to minimize the loss under worst-case adversarial perturbations. We find that overfitting to the training set does in fact harm robust performance to a very large degree in adversarially robust training across multiple datasets (SVHN, CIFAR-10, CIFAR-100, and ImageNet) and perturbation models (ℓ_∞ and ℓ_2). Based upon this observed effect, we show that the performance gains of virtually all recent algorithmic improvements upon adversarial training can be matched by simply using early stopping. We also show that effects such as the double descent curve *do* still occur in adversarially trained models, yet fail to explain the observed overfitting. Finally, we study several classical and modern deep learning remedies for overfitting, including regularization and data augmentation, and find that no approach in isolation improves significantly upon the gains achieved by early stopping. All code for reproducing the experiments as well as pretrained model weights and training logs can be found at https://github.com/locuslab/robust_overfitting.

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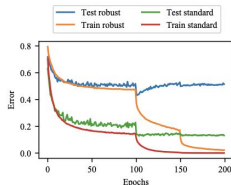


Figure 1. The learning curves for a robustly trained model replicating the experiment done by Madry et al. (2017) on CIFAR-10. The curves demonstrate “robust overfitting”; shortly after the first learning rate decay the model momentarily attains 43.2% robust error, and is actually more robust than the model at the end of training, which only attains 51.4% robust test error against a 10-step PGD adversary for ℓ_∞ radius of $\epsilon = 8/255$. The learning rate is decayed at 100 and 150 epochs.

1. Introduction

One of the surprising characteristics of deep learning is the relative *lack* of overfitting seen in practice (Zhang et al., 2016). Deep learning models can often be trained to zero training error, effectively memorizing the training set, seemingly without causing any detrimental effects on the generalization performance. This phenomenon has been widely studied both from the theoretical (Neyshabur et al., 2017) and empirical perspectives (Belkin et al., 2019), and remains such a hallmark of deep learning practice that it is often taken for granted.

In this paper, we consider the empirical question of overfitting in a similar, but slightly different domain: the setting of *adversarial training* for robust networks. Adversarial training is a method for hardening classifiers against adversarial attacks, i.e. small perturbations to the input which can drastically change a classifier’s predictions, that involves training the network on adversarially perturbed inputs instead of on clean data (Goodfellow et al., 2014). It is generally re-

2. Overfitting in Adversarially Robust Deep Learning

Reference: Rice, Leslie, et al. "Overfitting in Adversarially Robust Deep Learning."

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Definitions

What is Adversarial Training?

- It trains a model on images with small, deliberately added changes designed to cause misclassification. The goal is a model that resists these changes later.
- Normally in deep learning, training longer does not cause an AI model to perform worse. More training usually just makes a model better.
- This paper found that adversarial training contradicts that rule.

What is Robust Overfitting?

- The model keeps getting better at defending its training data, but gets worse at defending new data the longer training goes on.
- It shows up across many datasets and many types of attacks. It is not a one-time fluke.

Statistics

Title: Test Error Spikes Back Up

Subtitle: Robust error on CIFAR-10 across 200 training epochs

Chart: Train robust error (falls smoothly to near 0) vs. Test robust error (drops, then spikes right after epoch 100's learning rate decay, and keeps rising)

Key stat: 43.2% test robust error right after the learning rate drops, rising to 51.4% by the end of training

Training error keeps falling smoothly toward zero.

Test error drops, then spikes right when the learning rate decays, and never recovers

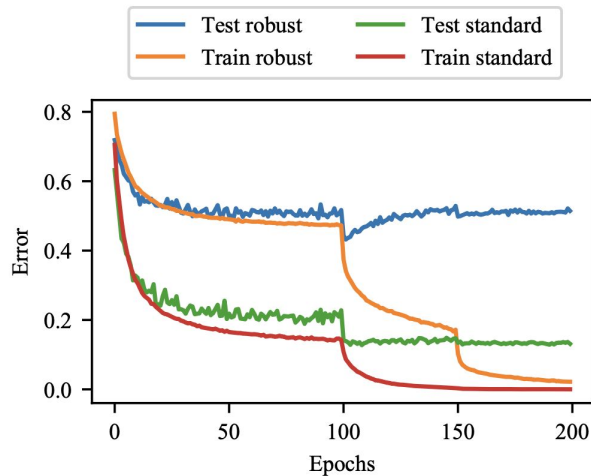


Figure 1. The learning curves for a robustly trained model replicating the experiment done by [Madry et al. \(2017\)](#) on CIFAR-10. The curves demonstrate “robust overfitting”; shortly after the first learning rate decay the model momentarily attains 43.2% robust error, and is actually more robust than the model at the end of training, which only attains 51.4% robust test error against a 10-step PGD adversary for ℓ_∞ radius of $\epsilon = 8/255$. The learning rate is decayed at 100 and 150 epochs.

Why Does This Happen?

- The timing is not a coincidence. Robust overfitting consistently begins right after the learning rate decays, across every schedule the authors tested.
- Smoother learning rate schedules (small gradual decreases instead of one sharp drop) do not fix the problem. They just spread the same overfitting pattern into a smoother curve. The single sharp decay schedule still gets the best peak performance.
- This looks like it should contradict "double descent," an idea from standard deep learning where bigger, more complex models keep getting better even after they perfectly fit the training data.
- The authors tested this directly by training models of many different sizes. Larger models did get better peak robust performance, matching double descent. But at every model size, training too long still caused the same overfitting and performance loss.
- Conclusion: model size and training length affect robust performance in separate, independent ways. Making the model bigger helps. Training it longer past the overfitting point hurts. One does not cancel out the other.

Early Stopping Method

Subtitle: Stop training right when test performance is best, instead of letting it run to the end.

Stat 1: Vanilla adversarial training on CIFAR-10 reaches 43.2% robust test error at its best early-stopped checkpoint, versus 51.4% if trained to the end. This early-stopped result matches TRADES' reported 43.4%, a far more complex method.

Stat 2: L2 adversarial training on CIFAR-10 goes from 31.1% (trained to the end) down to 28.4% (early stopped). Same pattern holds for a different attack type.

Stat 3: TRADES itself starts at 44.1% robust test error at its best early-stopped checkpoint, but degrades to 50.6% if trained to full convergence. Even complex methods rely on stopping early.

Bottom note: This is not the result of peeking at test data. A held-out validation set of just 1,000 examples, never seen during training, is enough to know exactly when to stop.

DATASET	NORM	RADIUS	ROBUST TEST ERROR (%)		
			FINAL	BEST	DIFF
SVHN	ℓ_∞	8/255	45.6 ± 0.40	39.0	6.6
	ℓ_2	128/255	26.4 ± 0.27	25.2	1.2
CIFAR-10	ℓ_∞	8/255	51.4 ± 0.41	43.2	8.2
	ℓ_2	128/255	31.1 ± 0.46	28.4	2.7
CIFAR-100	ℓ_∞	8/255	78.6 ± 0.39	71.9	6.7
	ℓ_2	128/255	62.5 ± 0.09	56.8	5.7
IMAGENET	ℓ_∞	4/255	85.5 ± 8.87	62.7	22.8
	ℓ_2	76/255	94.8 ± 1.16	63.0	31.8